Q1. What is the differential diagnosis of ulcers over the penis? Describe the clinical features and a treatment of syphilis.

Q2. Blood-Borne Infections

Q3. Describe the clinical features, investigations and treatment of thyphoid fever.

Q4. Describe the etiology, clinical features and management of Diphtheria.

Q5. Clinical features and treatment of H1N1 Influenza (Seasonal Flu)

Q6. Oral manifestation of HIV/AIDS

Q7. RNTCP (Revised National Tuberculosis Control Program)

Q8. Dengue Fever

Q9. Herpes Simplex

Q10. Herpes Zoster

Q11. Mumps

Q12. Malaria

Q13. Clinical features, diagnosis and management of P. falciparum Malaria

Q14. Dengue Hemorrhagic Fever (DHF)

Q15. Cerebral malaria.

Q1. What is the differential diagnosis of ulcers over the penis? Describe the clinical features and a treatment of syphilis.

Differential Diagnosis of Ulcers Over Penis

- **Chancre**
- **Chancroid** (painful, non-indurated ulcer)
- **Lymphogranuloma venereum**
- **Genital herpes**
- **Neoplasm**

Syphilis
Aetiopathogenesis

- Caused by *Treponema pallidum*.

- Transmitted via sexual contact or transplacentally.
- Chronic systemic infection with latent periods.

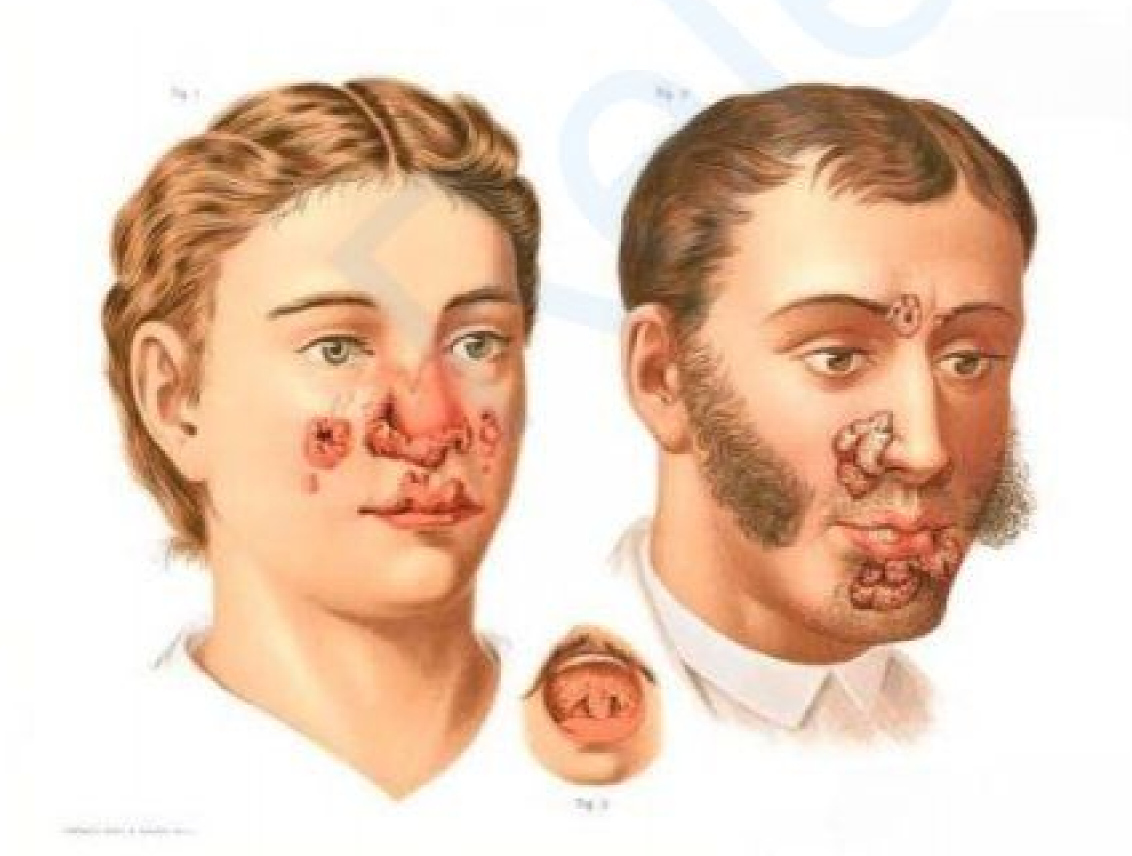
Clinical Features
Primary Syphilis

- Chancre: Single, reddish, indurated, painless lesion at the site of inoculation (e.g., penis in heterosexual males).

Secondary Syphilis

- Develops 6-8 weeks after primary infection.
- Skin lesions: Macular, papular, or pustular.
- **Condyloma lata**: Found in genital/anal regions.
- **Mucous patches**: Highly infectious.
- Other features: Alopecia, iritis, hearing loss, arthritis, nephritis, hepatitis, meningitis.

Tertiary Syphilis



- **Gumma**: Affects skin, mucous membranes, bones.
- CVS: Aortitis, aortic aneurysms, aortic regurgitation, coronary ostial stenosis, gummatous myocarditis.
- **Neurosyphilis**: Meningeal, cerebrovascular syphilis, tabes dorsalis, general paresis.

Congenital Syphilis

- **Early**: Snuffles, failure to thrive, condyloma lata, osteochondritis, periostitis.
- **Late**: Interstitial keratitis, neurosyphilis, sensorineural deafness, gummas, bony involvement.
- **Stigmata**: Rhagades, mulberry molars, Hutchinson teeth, frontal bossing, saddle nose, saber tibia.
- **Hutchinson's Triad**: Hutchinson teeth, interstitial keratitis, eighth nerve deafness.

Diagnosis

- Single painless lesion with defined edges.
- **Dark field microscopy**
- **VDRL test**
- **TPHA**
- Heals in 2-6 weeks without treatment.

Treatment

- **Primary, Secondary, Latent Syphilis**: Benzathine penicillin 24 lakh units IM stat.
- **Tertiary Syphilis**: Benzathine penicillin 24 lakh units IM weekly for 3 weeks.
- **Alternatives**: Doxycycline, erythromycin, azithromycin, ceftriaxone.

Q2. Blood-Borne Infections

Definition

- Diseases transmitted through **blood** or **body fluids** containing blood.

Modes of Transmission

- **Blood transfusion**
- **Needle-stick injuries**
- **Sharing needles** (e.g., IV drug users)
- **Sexual contact**
- **From mother to child** (vertical transmission).

Common Blood-Borne Infections

1. **Hepatitis B (HBV)**
    - Caused by **Hepatitis B virus**.
    - **Symptoms**: Fatigue, jaundice, abdominal pain, dark urine, nausea.
    - Can cause **chronic liver disease** and **hepatocellular carcinoma**.
2. **Hepatitis C (HCV)**
    - Caused by **Hepatitis C virus**.
    - **Symptoms**: Mostly asymptomatic, but may cause **chronic liver damage**.
    - Leads to **cirrhosis** or **liver cancer**.
3. **HIV/AIDS**
    - Caused by **Human Immunodeficiency Virus**.
    - Weakens **immune system** by attacking CD4 cells.
    - Leads to **opportunistic infections** and cancers in AIDS stage.
4. **Syphilis**
    - Caused by **Treponema pallidum**.
    - Transmitted via **blood** or **sexual contact**.
5. **Malaria**
    - Caused by **Plasmodium species**.
    - Transmitted through infected **blood transfusion** or **needle sharing**.

Prevention

- Use of **disposable needles**.
- Screening of **blood donors**.
- **Safe sex practices** (use of condoms).
- **Vaccination** (e.g., for HBV).
- **Proper sterilization** of medical equipment.

Treatment

- **HBV**: Antiviral medications like **tenofovir** or **entecavir**.
- **HCV**: Direct-acting antivirals (DAAs) like **sofosbuvir**.
- **HIV/AIDS**: Antiretroviral therapy (ART).
- **Malaria**: Antimalarial drugs like **artemisinin** or **chloroquine**.

Q3. Describe the clinical features, investigations and treatment of thyphoid fever.

Typhoid Fever (Enteric Fever)

Definition

- Acute systemic illness caused by **Salmonella typhi** and **S. paratyphi**.

Epidemiology

- **Reservoir**: Acutely ill, convalescing patients, chronic carriers.
- **Transmission**: Faeco-oral route.
- Common in **children** and **young adults**.

Aetiopathogenesis Typhoid

- Invasion of **Peyer's patches**, causing **transient bacteraemia**.
- Persistent bacteraemia due to **phagocytosis** by reticuloendothelial system.
- Typical lesion: **Hyperplasia, necrosis, and ulceration** in Peyer's patches.
- Systemic effects due to **toxaemia** and **septicaemia**.

Clinical Features

- **Onset**: Insidious, with an incubation period of **10–14 days**.
- **1st week**:
    - **Prodromal symptoms**: Remittent fever, headache, myalgia, malaise, constipation, leucopenia, relative bradycardia.
    - Children: **Diarrhoea** and **vomiting**.
    - Fever rises in a **stepwise pattern** and reaches a plateau.
- **2nd week**:
    - **Rose spots**, splenomegaly, bronchitis, abdominal pain, distention, diarrhoea.
- **Beyond 2nd week**:
    - **Confusion**, delirium, complications, coma, or death.
- **Relapse**: Symptoms reappear within **2 weeks** of recovery.
- **Chronic carrier state**:
    - Seen in **5% of cases** (commonly females).
    - Due to **infection persistence** in the gallbladder.

Complications

- **Intestinal**: Haemorrhage, paralytic ileus, perforation, peritonitis.
- **Extraintestinal**:
    - Meningitis, cholecystitis, pneumonia, myocarditis.
    - Bone/joint infections, encephalopathy, granulomatous hepatitis, nephritis.

Treatment

1. **Antibiotics** (oral or parenteral based on condition):
    - **Fluoroquinolones**
    - **Third-generation cephalosporins**: Ceftriaxone, cefotaxime, cefixime, cefpodoxime.
    - **Azithromycin**
2. **Commonly used antibiotics**:
    - **Ciprofloxacin**: 500 mg BID for 10 days.
    - **Ofloxacin**: 400 mg BID for 7 days.
    - **Ceftriaxone**: 1 g IV BID for 10 days.
    - **Azithromycin**: 1 g daily for 5 days.
3. **Severe cases**:
    - **Dexamethasone**: 3 mg/kg (single dose), followed by 1 mg/kg every 6 hours (8 doses).
4. **Carriers**: Oral antibiotics (**Amoxicillin, Ciprofloxacin, TMP-SMX**) for 6 weeks.

Q4. Describe the etiology, clinical features and management of Diphtheria.

Cause

- Caused by **Corynebacterium diphtheriae**, a gram-positive rod with **Chinese letter configuration**.
- Produces a **powerful exotoxin** damaging the **heart muscle** and **nervous system**.

AetiopathogenesisDiphtheria

- Humans are the only **reservoir**.
- **Transmission**: Aerosol inhalation and direct contact.
- **Diphtheria toxin** causes **tissue necrosis**.

Clinical Features

- **Incubation period**: 2–4 days (average 1 week).
- Forms: **Nasal, pharyngeal (most common), laryngeal, cutaneous**.
- **Pharyngeal form**:
    - Low-grade fever, **pharyngeal erythema**, and **pseudomembrane** formation.
- **Laryngeal involvement**: Hoarseness, stridor, **respiratory obstruction.**
- **Nasal diphtheria**: Nasal discharge.
- Other symptoms: **Otitis media, conjunctivitis, keratitis, myocarditis, neuropathies**.
- **Severe cases**: May cause **acute circulatory failure** and death.

Diagnosis

- **Gram's stain**
- **Bacterial culture**
- Clinical identification of **pseudomembrane** and isolation of *C. diphtheriae*.

Treatment

1. **Isolation of patient**.
2. **Antitoxin administration**: Neutralizes free toxin.
   - **Mild cases**: 4,000–8,000 units IM.
   - **Moderate cases**: 16,000–40,000 units IM.
   - **Severe cases**: Up to 100,000 units IV.
3. **Antibiotics**: Prevents spread of *C. diphtheriae*.
   - **Amoxicillin**: 500 mg 8th hourly for 10–14 days.
   - **Erythromycin**: 500 mg 6th hourly.
   - **Clarithromycin, Azithromycin**: Equally effective.
4. **Supportive measures**:
   - **Tracheostomy** in respiratory obstruction.
   - **Immunization** with diphtheria toxoid after recovery.

Prevention

- **Contact tracing, surveillance, prophylactic antibiotics**.
- Routine **immunization**.

Complications

1. **Cardiac**:
   - **Toxic myocarditis**: Tachycardia, feeble heart sounds, cardiac enlargement, tic-tac rhythm, arrhythmias.
   - Sudden death from **congestive heart failure**.
2. **Respiratory**:
   - Bronchitis, bronchopneumonia, **respiratory obstruction**, **respiratory paralysis**.
3. **Neurological**:
   - **Paralysis** of palate, accommodation, facial muscles, bulbar muscles, respiratory muscles.
4. **Renal**:
   - **Toxic nephritis**.
5. **Vascular**:
   - **Middle cerebral artery thrombosis**, monoplegia.
6. Other: **Otitis media**, **arthritis**.

Q5. Clinical features and treatment of H1N1 Influenza (Seasonal Flu)

H1N1 Influenza

- **H1N1 influenza** is a type of **flu** caused by the H1N1 virus.
- The virus is part of the **influenza A** family, which is known to affect the **respiratory system**.
- **Transmission** occurs through **droplets** when an infected person coughs, sneezes, or talks.
- **Symptoms** include:
  - **Fever**
  - **Cough**
  - **Sore throat**
  - **Body aches**
  - **Fatigue**
- **Risk factors** include:
  - **Young children**
  - **Elderly**
  - **Pregnant women**
  - **People with weak immune systems**

Clinical Features

1. **Incubation period**: 1–4 days.
2. **General symptoms**:
   - **Sudden onset** fever, chills.
   - **Fatigue**, weakness, malaise.
   - **Headache** and body ache (myalgia).
3. **Respiratory symptoms**:
   - **Dry cough** (can progress to productive).
   - **Sore throat**, nasal congestion, runny nose.
   - **Shortness of breath** in severe cases.
4. **Gastrointestinal symptoms** (in some cases):
   - Nausea, vomiting, diarrhea.
5. **Severe cases**:
   - **Pneumonia**, respiratory failure, multi-organ dysfunction.
   - Can lead to **death** in high-risk patients (pregnant women, elderly, immunocompromised individuals).

Treatment

1. **Antiviral medications**:
   - **Oseltamivir (Tamiflu):** 75 mg BID for 5 days.
   - **Zanamivir**: Inhalation, 10 mg BID for 5 days.
2. **Supportive care**:
   - **Rest** and **hydration**.

    - Paracetamol for **fever** and body aches.
    - Avoid aspirin (risk of **Reye's syndrome**).
3. **Hospitalization** for severe cases:
    - Oxygen therapy or mechanical ventilation in **respiratory distress**.
    - IV fluids for dehydration.
4. **Prevention**:
    - **Annual flu vaccination**.
    - Personal hygiene: Hand washing, wearing masks.
    - Isolation of infected individuals.
5. Diagnosis is done through **PCR testing** and **rapid antigen tests**.

Q6. Oral manifestation of HIV/AIDS

- **Global pandemic** caused by **Human Immunodeficiency Virus (HIV)**, a retrovirus of the **Lentivirus sub-family**.
- Leads to **Acquired Immunodeficiency Syndrome (AIDS)**.

Oral Manifestations of AIDS

- **Oropharyngeal lesions** are common and result from **secondary infections**.

1. Fungal Infections

- **Candidiasis** (most common)
- **Histoplasmosis**
- **Cryptococcosis**

2. Viral Infections

- **Oral hairy leukoplakia** (caused by EBV)
- **Herpes simplex**
- **Cytomegalovirus**
- **Varicella zoster**
- **Papilloma virus infections**

3. Bacterial Infections

- **Periodontal infections**
- **Necrotizing ulcerative gingivitis**
- **Necrotizing stomatitis**

4. Neoplasms

- **Kaposi sarcoma**
- **Lymphoma**

5. Idiopathic Lesions

- **Aphthous ulcers**

6. HIV Salivary Gland Disease (DILS)

7. Cervical Lymphadenopathy

Key Oral Lesions

- **Candidiasis (Thrush)**: Indicator of underlying HIV. Appears as **white, cheesy exudates** on erythematous mucosa in the **posterior oropharynx**.
- **Ulcers**: Palatal, glossal, or gingival ulcers due to **cryptococcal disease** or **histoplasmosis**.

These oral manifestations can aid in the **early diagnosis** and monitoring of **HIV progression**.

Q7. RNTCP (Revised National Tuberculosis Control Program)

Introduction

- **Launched** in 1997 in India to combat **tuberculosis (TB)**.
- Based on the **DOTS (Directly Observed Treatment Short-course)** strategy recommended by WHO.
- Aims to achieve **universal access to TB care**.

Goals

- **Detect at least 70%** of infectious TB cases.
- Ensure **85% cure rate** for infectious TB cases.
- Prevent the **development of drug resistance**.

Components of RNTCP

1. **Political and administrative commitment**:
    - Ensures resources and infrastructure for TB control.
2. **Case detection through sputum microscopy**:
    - Free and accessible diagnostic services.
3. **Standardized treatment regimens**:
    - Categorized treatment based on the type and severity of TB.
4. **Directly observed treatment (DOT)**:
    - Health workers ensure the patient takes medicine.
5. **Drug supply**:
    - Uninterrupted and free supply of quality drugs.
6. **Monitoring and supervision**:

- Regular recording and reporting of TB cases.

Categories of Treatment

1. **Category I**:
   - **New cases of pulmonary TB.**
   - **Regimen**: 2 months of intensive phase (HRZE) + 4 months continuation (HR).
2. **Category II**:
   - **Retreatment cases** (relapse, treatment failure, default).
   - **Regimen**: 2 months of intensive phase (HRZES) + 1 month (HRZE) + 5 months continuation (HRE).
3. **Category III**:
   - **Extrapulmonary TB** or less severe cases.

DOTS Strategy

- Ensures **completion of treatment** under supervision.
- Reduces **default rates** and prevents **drug resistance**.

Achievements of RNTCP

- Reduction in TB **morbidity and mortality**.
- Increased **case detection rates**.
- Expanded access to **diagnostic and treatment services**.

Challenges

- Rise of **MDR-TB (Multidrug-resistant TB)**.
- Lack of awareness among patients.
- **Stigma** associated with TB.

Transition to NTEP

- In 2020, RNTCP was renamed **National Tuberculosis Elimination Program (NTEP)** with the goal of **eliminating TB by 2025**.

Conclusion

- RNTCP has been a **landmark program** in controlling TB in India, with a focus on ensuring effective treatment and reducing disease burden.

Q8. Dengue Fever

Introduction

- Dengue is a **viral infection** caused by the **Dengue virus**.
- Transmitted by the **Aedes aegypti mosquito**.
- Common in tropical and subtropical regions, especially during the **rainy season**.

Aetiology

- Caused by **Dengue virus** (flavivirus), which has **four serotypes**: DEN-1, DEN-2, DEN-3, and DEN-4.
- Infection by one serotype provides immunity only to that serotype, not the others.

Pathogenesis

- The virus enters the **bloodstream** after mosquito bite and infects **monocytes, macrophages**, and **dendritic cells**.
- Immune response and **cytokine release** lead to **vascular leakage**, causing **plasma loss** and **hemoconcentration**.
- This leads to **fever**, **shock**, and **bleeding tendencies**.

Clinical Features

1. **Incubation Period**: 4–10 days after mosquito bite.
2. **Acute Phase**:
    - **High fever** (up to 104°F), sudden onset.
    - **Severe headache**, **retro-orbital pain**.
    - **Muscle and joint pain** (dengue aches).
    - **Skin rash** (appears after 3–4 days, may be petechial).
    - **Malaise**, **fatigue**, and **loss of appetite**.
3. **Hemorrhagic manifestations**:
    - **Petechiae**, **purpura**, **epistaxis**, **gum bleeding**.
4. **Severe cases** (Dengue Hemorrhagic Fever - DHF):
    - **Plasma leakage**, **shock**, **bleeding**.
    - **Organ failure** and **death** can occur in severe cases.
5. **Recovery phase**:
    - Recovery typically begins after 2-7 days, but weakness and fatigue may persist.

Classification of Dengue

1. **Classical Dengue**: Mild illness with fever, rash, and minor bleeding.
2. **Dengue Hemorrhagic Fever (DHF)**: Includes bleeding, **plasma leakage**, and **shock**.
3. **Dengue Shock Syndrome (DSS)**: Severe shock due to **plasma leakage** and **vascular collapse**.

Investigations

1. **Clinical diagnosis** based on symptoms and **epidemiological history** (recent travel to endemic areas).
2. **Laboratory Tests**:
   - **Complete Blood Count (CBC)**:
     - **Leukopenia** (low white blood cells).
     - **Thrombocytopenia** (low platelets).
     - **Hemoconcentration** (raised hematocrit).
   - **Dengue NS1 Antigen Test**: Detects **NS1 antigen** in the early phase (1–5 days of illness).
   - **Dengue IgM and IgG Antibody Test**:
     - IgM is positive 3–5 days after infection (indicates acute infection).
     - IgG appears later (indicates past infection).
   - **PCR (Polymerase Chain Reaction)**: Can detect viral RNA in the early stages.
   - **Liver Function Tests**: Elevated **transaminases** (AST, ALT).
3. **Coagulation Profile**:
   - **Prolonged PT** (prothrombin time), **aPTT** in severe cases.
   - **Decreased fibrinogen** and **increased D-dimer** may indicate severe disease.

Treatment

1. **Supportive Care**:
   - **Hydration** is the mainstay of treatment.
   - Oral rehydration for mild cases, intravenous fluids for severe cases.
   - **Paracetamol** for fever and pain.
   - Avoid **aspirin** and **NSAIDs** due to the risk of bleeding.
2. **Platelet Transfusion**: Indicated in cases with **severe thrombocytopenia** and bleeding.
3. **Blood Pressure Monitoring**: For patients in shock (DSS), **IV fluids** and **vasopressors** may be required.
4. **Oxygen therapy**: For patients with respiratory distress due to **shock** or **organ failure**.

Prevention

1. **Vector Control**:
   - Eliminate **mosquito breeding sites** (stagnant water).
   - Use **mosquito repellents**, **nets**, and **wear long-sleeved clothing**.
   - **Insecticide spraying** in endemic areas.
2. **Vaccination**:
   - The **Dengue vaccine (CYD-TDV)** is available but recommended for individuals previously infected with dengue.

Prognosis

- Most cases of dengue are **self-limiting** with recovery in 1–2 weeks.
- Severe cases like **Dengue Hemorrhagic Fever** or **Dengue Shock Syndrome** may have a **high mortality rate** if not managed properly.

Conclusion

- Dengue is a **serious disease** with varying degrees of severity.
- Early detection, supportive care, and timely intervention can significantly reduce the risk of complications and death.

Q9. Herpes Simplex

Epidemiology and Pathogenesis

- Caused by **Herpes Simplex Virus (HSV)**, which has two types:
  - **HSV-1**: Causes **cutaneous, oropharyngeal**, and **ocular** lesions.
  - **HSV-2**: Causes **genital herpes**.
- Primary infection is often **asymptomatic** and occurs commonly in **childhood** via **mucosal surfaces** or **abraded skin**.
- After initial infection, the virus remains **dormant** in **sensory** and **autonomic ganglia**.
- **Reactivation** can be triggered by factors like **stress**, **trauma**, **menstruation**, and **fever**.

Clinical Features

1. **Orofacial and Eye Infections**:
   - **Ulcerative gingivostomatitis**: Painful ulcers in the mouth.
   - **Pharyngitis**: Inflammation of the throat.
   - **Herpes labialis**: Cold sores around the lips.
   - **Keratoconjunctivitis**: Inflammation of the cornea and conjunctiva.
2. **Genital Infections**:
   - **Vulvovaginitis**: Inflammation of the vulva and vagina.
   - **Urethritis**: Inflammation of the urethra.
   - **Proctitis**: Inflammation of the rectum.
3. **Nervous System Infections**:
   - **Encephalitis**: Inflammation of the brain.
   - **Guillain-Barré syndrome**: A condition where the immune system attacks the peripheral nervous system.
   - **Transverse myelitis**: Inflammation of the spinal cord.
4. **Visceral Infections**:
   - **Oesophagitis**: Inflammation of the esophagus.

- **Pneumonitis**: Inflammation of the lungs.
- **Hepatitis**: Inflammation of the liver.

Diagnosis

- The **characteristic feature** of herpes simplex infection is **multiple vesicles** (blisters) over an **erythematous base** (red, inflamed skin).
- **Laboratory confirmation** can be done by:
  - **Tzanck test** (looking for multinucleated giant cells).
  - **Tissue culture**: Growing the virus from a sample.
  - **Serology**: Detecting antibodies against the virus.
  - **PCR (Polymerase Chain Reaction)** for **HSV DNA** in **CSF** (cerebrospinal fluid) in cases of encephalitis.

Treatment

1. **Mucocutaneous Infections**:
   - **Acyclovir**, **Famciclovir**, and **Valacyclovir** are used to treat these infections.
2. **Eye Infections**:
   - **Idoxuridine**, **Vidarabine**, and **Cidofovir** are used for eye infections.
3. **HSV Encephalitis**:
   - **Intravenous acyclovir** (10 mg/kg every 8 hours for 10 days) is the treatment for **HSV encephalitis**.

Conclusion

- Herpes simplex is a **common viral infection** that can cause a variety of **skin, mucosal**, and **nervous system** infections.
- **Early diagnosis** and **appropriate antiviral treatment** can help manage symptoms and prevent complications.

Q10. Herpes Zoster

- Caused by **Varicella Zoster Virus (VZV)**, the same virus responsible for **chickenpox**.
- **Primary infection** is **chickenpox**, usually acquired during childhood.
- The virus remains dormant in **dorsal root** and **cranial nerve ganglia**. Reactivation of the virus causes **herpes zoster** (shingles).
- Common in **middle-aged** or **elderly** individuals.

Clinical Features

- **Vesicular eruption** in a **dermatomal pattern** (usually on one side of the body).
- **Severe pain** usually precedes the rash by **2 to 3 days**.

- **Ramsay-Hunt syndrome**: Occurs due to the involvement of the **geniculate ganglion** of the **facial nerve**, leading to **facial paralysis** and **vesicular rash** in the ear or mouth.

Complications

- **Pneumonitis**: Inflammation of the lungs, especially in immunocompromised patients.
- **Encephalomyelitis**: Inflammation of the brain and spinal cord.
- **Hepatitis**: Inflammation of the liver.
- **Post-herpetic neuralgia**: Persistent nerve pain after the rash has healed.

Diagnosis

- **Clinical presentation**: Typical vesicular rash in a dermatomal distribution.
- **PCR** and **Tzanck test** can confirm the presence of VZV DNA.

Treatment

- **Acyclovir**: 800 mg, 5 times a day for **7 to 10 days**.
- Local treatments:
    - **Calamine lotion** for soothing the skin.
    - **Antiseptic powders** to prevent secondary infections.

For Post-Herpetic Neuralgia (pain that persists after rash resolution):

- **Analgesics** for pain relief.
- **Amitriptyline**: A tricyclic antidepressant used for nerve pain.
- **Carbamazepine**: An anticonvulsant that can help with nerve pain.
- **Sodium valproate**: Used in cases of severe neuralgia.
- **Stellate ganglion block**: A nerve block technique to relieve pain.
- **Electrical nerve stimulation**: To reduce nerve pain.

Key Points

- Herpes zoster is caused by the reactivation of the varicella zoster virus.
- It presents with a **painful rash** in a dermatomal pattern.
- **Early antiviral treatment** can reduce the severity and duration of the infection.
- Management of **post-herpetic neuralgia** is crucial to alleviate long-term pain.

Q11. Mumps

Caused by:

- **Paramyxovirus**

Mode of Spread:

- Spread by **droplet infection** (coughing, sneezing).

Incubation Period:

- **15-20 days**

Clinical Features:

1. **Fever**: Often **high fever** occurs at the onset.
2. **Headache**: Common in the early stages of the infection.
3. **Vomiting**: Occurs due to the general systemic infection.
4. **Anorexia**: Loss of appetite is a common symptom.
5. **Malaise**: General feeling of **discomfort** and **weakness**.
6. **Parotid Enlargement**: The **parotid glands** (salivary glands) become swollen, typically the most noticeable feature of mumps. This leads to **pain** and **swelling** near the jaw and neck area.
7. **Orchitis**: **Inflammation of the testes**, which is a classic complication. It may lead to **testicular atrophy** (shrinkage) if untreated.
8. **Neurological Complications**:
   - **Aseptic meningitis**: Inflammation of the protective membranes around the brain and spinal cord.
   - **Meningoencephalitis**: Inflammation of both the brain and its protective coverings.
   - **Neuritis**: Inflammation of the nerves.
   - **Myelitis**: Inflammation of the spinal cord.
9. **Other Complications**:
   - **Arthritis**: Joint inflammation.
   - **Myocarditis**: Inflammation of the heart muscle.
   - **Hepatitis**: Inflammation of the liver.
   - **Involvement of other glands**: Other salivary glands, pancreas, or reproductive organs may also be affected.

Diagnosis:

- **Isolation of the virus** from throat swabs: Detecting the virus from patient samples.
- **Serology**: Detection of antibodies to the mumps virus in the blood.
- **Elevated serum amylase levels**: High levels of **amylase** (an enzyme) are often found in blood tests, indicating salivary gland involvement.

Treatment:

- **Supportive care** is the main treatment approach. This includes:

- **Rest** and **hydration**.
    - **Pain relievers** such as acetaminophen or ibuprofen.
    - **Warm or cold compresses** on the swollen glands to relieve discomfort.
    - **Anti-inflammatory drugs** may be used for complications like orchitis or arthritis.

Prevention:

- **Attenuated live mumps vaccine** is used for prevention.
    - Typically administered at **18 months** of age as part of the **MMR vaccine** (measles, mumps, rubella).

Conclusion:

- Mumps is a **viral infection** that primarily affects the **parotid glands** and can lead to severe complications such as **orchitis**, **aseptic meningitis**, and other organ involvements.
- **Vaccination** remains the most effective method of preventing the disease, with early supportive treatment for symptomatic relief and management of complications.

Q12. Malaria

Caused by:

- **Protozoan disease** transmitted by the bite of **Anopheles mosquitoes**.
- The **Plasmodium** species cause malaria:
    - **P. vivax**
    - **P. falciparum**
    - **P. malariae**
    - **P. ovale**

Clinical Features:

- **Malaise**: General feeling of discomfort and unwellness.
- **Fatigue**: Extreme tiredness, a hallmark of the disease.
- **Headache**: A common symptom, often severe.
- **Fever**: The fever pattern is **cyclical**, often occurring every 48 hours (for P. vivax and P. ovale) or 72 hours (for P. malariae).
- **Chills and Rigors**: Severe shaking chills followed by high fever.
- **Splenomegaly**: **Enlargement of the spleen** due to the body's attempt to filter the infected red blood cells.
- **Relapses**: Seen specifically in **P. vivax** malaria, as dormant liver stages (hypnozoites) can cause reactivation of the infection months after initial treatment.

- **Tropical Splenomegaly Syndrome**: Chronic **splenomegaly** and **anemia** seen in endemic areas.

Complications (More common in P. falciparum infections):

- **Cerebral Malaria**: A severe form of malaria causing **neurological symptoms**, including confusion, coma, and seizures.
- **Severe Anaemia**: Due to the destruction of red blood cells infected by the parasite.
- **Acute Renal Failure**: Caused by severe malaria, especially in cases of **hemolysis** (breakdown of red blood cells).
- **ARDS** (Acute Respiratory Distress Syndrome): A severe lung condition leading to difficulty breathing and low oxygen levels.
- **Hypoglycemia**: Low blood sugar, which can be fatal if untreated.
- **Shock**: Severe **hypotension** (low blood pressure) due to sepsis and severe anemia.
- **DIC** (Disseminated Intravascular Coagulation): A clotting disorder where small clots form throughout the body.
- **Seizures**: Due to **cerebral malaria** or severe infection.

Diagnosis:

1. **Peripheral Smear**:
    - **Thick smear**: Used to detect the presence of **Plasmodium** parasites.
    - **Thin smear**: Used for identifying the species of **Plasmodium** by examining the shape and size of the parasites.
2. **Quantitative Buffy Coat Analysis**: A method of **concentrating the blood sample** to detect the parasite load.
3. **Rapid Diagnostic Tests (RDTs)**: Detects antigens from the malaria parasite in the blood.
4. **PCR (Polymerase Chain Reaction)**: Used in specialized labs to confirm the type of **Plasmodium** and detect **low levels** of the parasite.

Treatment:

1. **For P. vivax, P. malariae, P. ovale, and uncomplicated P. falciparum malaria**:
    - **Oral chloroquine**:
        - **600 mg** initially, followed by **300 mg** after 6, 24, and 48 hours.
        - **For P. vivax**, **primaquine** may also be used to eliminate dormant liver forms.
2. **For complicated P. falciparum malaria (severe cases)**:
    - **Intravenous quinine**:
        - **10 mg/kg** over **4-8 hours**, thrice daily for **7 days**.
    - **Artesunate**:
        - **2 mg/kg** initially (stat dose), followed by **1 mg/kg daily** for **4 days**.
3. **Supportive Treatment**:

- **Intravenous fluids** and **electrolyte management** for dehydration.
- **Blood transfusion** in cases of severe anemia.
- **Antipyretics** (like **paracetamol**) to control fever.

Prevention:

- **Mosquito control**: Use of **insecticide-treated nets** and **indoor spraying** with insecticides.
- **Prophylactic medications**: For travelers to malaria-endemic areas, **malarial prophylaxis** with **chloroquine** or **malarone**.
- **Vaccines**: Currently, the **RTS,S/AS01 vaccine** (Mosquirix) is in use in some areas for children.

Conclusion:

- **Malaria** is a serious, life-threatening disease caused by **Plasmodium** parasites transmitted by **Anopheles mosquitoes**.
- Early diagnosis and treatment are crucial, with **chloroquine** and **quinine** being the mainstay of treatment. Severe cases require urgent medical intervention to manage complications like **cerebral malaria** and **renal failure**.

Q13. Clinical features, diagnosis and management of P. falciparum Malaria

Clinical Features:

- **High-grade fever** with chills and rigors.
- **Headache** and severe **malaise**.
- **Nausea, vomiting**, and diarrhea in some cases.
- **Splenomegaly** (enlarged spleen).
- **Jaundice** due to hemolysis of infected red blood cells.
- **Cerebral malaria**: Confusion, altered consciousness, seizures, or coma.
- **Severe anemia**: Rapid destruction of red blood cells.
- **Acute respiratory distress syndrome (ARDS)**: Difficulty in breathing.
- **Hypoglycemia** due to parasite metabolism and drug side effects.
- **Acute renal failure** caused by hemoglobinuria ("blackwater fever").
- **Shock** and hypotension in severe cases.
- **Disseminated intravascular coagulation (DIC)** leading to bleeding complications.

Diagnosis:

- **Peripheral blood smear**:
  - **Thick smear**: Detects malaria parasites.
  - **Thin smear**: Differentiates between Plasmodium species.
- **Rapid Diagnostic Tests (RDTs)**: Detects malaria antigens in the blood.

- **Quantitative Buffy Coat Analysis**: Identifies parasitemia levels.
- **PCR (Polymerase Chain Reaction)**: Confirms species and detects low-level parasitemia.
- **Liver and kidney function tests**: Check for complications.
- **Blood glucose levels**: Monitors for hypoglycemia.

Management:

- **First-line treatment for uncomplicated falciparum malaria**:
    - **Artemisinin-based combination therapy (ACT)** (e.g., Artesunate + Lumefantrine).
    - **Chloroquine is ineffective** due to resistance.
- **Severe falciparum malaria**:
    - **Intravenous artesunate**: 2.4 mg/kg stat, followed by 2.4 mg/kg at 12 and 24 hours, then daily for 7 days.
    - **Intravenous quinine**: 10 mg/kg every 8 hours for 7 days.
- **Supportive care**:
    - **Intravenous fluids** for dehydration and shock.
    - **Blood transfusion** in cases of severe anemia.
    - **Oxygen therapy** and ventilation support in ARDS.
    - **Antipyretics** like paracetamol for fever.
- **Monitor complications**: Regular check of blood glucose, kidney function, and neurological status.

Prevention:

- Use **insecticide-treated nets** and **indoor residual spraying**.
- **Prophylactic antimalarial drugs** (e.g., doxycycline) for travelers.
- Early treatment of symptoms to prevent severe complications.

Key Points:

- **P. falciparum malaria** is the most severe form of malaria with high morbidity and mortality.
- Early diagnosis with **smear** or **RDT** and prompt treatment with **ACT** is essential to improve outcomes.
- Severe cases require **intensive care** to manage complications effectively.

Q14. Dengue Hemorrhagic Fever (DHF)

Etiopathogenesis:

- **Caused by Dengue virus**: A **flavivirus** transmitted by the **Aedes mosquito**.
- **Four serotypes** of Dengue virus (DEN-1, DEN-2, DEN-3, DEN-4).

- **Primary infection**: Typically results in **mild dengue fever**.
- **Secondary infection**: **Dengue Hemorrhagic Fever** occurs when a person is infected with a **different serotype** than the first, causing a **more severe immune response**.
- **Viral replication**: Occurs in **macrophages**, **dendritic cells**, and **endothelial cells**.
- **Cytokine storm**: Leads to **vascular leakage**, causing **plasma loss** into the extravascular space.
- **Increased capillary permeability**: Results in **edema**, **hypovolemia**, and **hemoconcentration**.
- **Platelet dysfunction** and **coagulation abnormalities** lead to **bleeding**.

Clinical Features:

- **Fever**: Sudden onset of **high fever**, typically lasting 2-7 days.
- **Severe headache** and **retro-orbital pain**.
- **Muscle and joint pain** (myalgia and arthralgia).
- **Rash**: Initially **maculopapular**, may progress to **petechiae** and **ecchymosis**.
- **Bleeding manifestations**:
    - **Mucosal bleeding**: **Gum bleeding**, **epistaxis**.
    - **Hematochezia**, **hematuria**, and **petechiae**.
    - **Severe hemorrhage** in the skin, gastrointestinal tract, or internal organs.
- **Shock**: Due to **plasma leakage**, leading to **hypovolemia** and **hypotension** (severe DHF).
- **Abdominal pain** and **hepatomegaly** (liver enlargement).
- **Complications**:
    - **Severe thrombocytopenia** (<100,000 platelets).
    - **Plasma leakage** and **shock** (dengue shock syndrome, DSS).
    - **Organ failure** (liver, heart, and kidneys).

Diagnosis:

- **Clinical criteria**: Fever, hemorrhagic manifestations, and thrombocytopenia in a high-risk area.
- **Laboratory tests**:
    - **Complete blood count (CBC)**: **Thrombocytopenia** (<100,000 platelets) and **leukopenia**.
    - **Hematocrit**: Elevated due to **hemoconcentration** (plasma leakage).
    - **Serology**: Detection of **IgM** (recent infection) and **IgG** (past infection) antibodies.
    - **Polymerase chain reaction (PCR)**: Detects **Dengue virus RNA**.
    - **NS1 antigen**: Detected in the first few days of infection.
    - **Liver function tests**: May show elevated **AST/ALT**.
    - **Coagulation profile**: May show **prolonged PT** and **aPTT** due to coagulopathy.

Management:

- **Supportive treatment**:
  - **Fluid management** is the cornerstone of treatment.
    - **Oral rehydration** for mild cases.
    - **Intravenous fluids** (isotonic saline or Ringer's lactate) for moderate to severe cases to correct **hypovolemia** and maintain **circulatory volume**.
    - **Plasma expanders** (e.g., albumin) may be used in severe cases.
  - **Monitoring**: Continuous monitoring of **vital signs**, **hematocrit**, and **platelet count**.
  - **Oxygen therapy** if the patient shows signs of **respiratory distress** or **shock**.
- **Antipyretics**:
  - **Paracetamol** (acetaminophen) is used for fever management.
  - **Aspirin and NSAIDs** should be **avoided** due to the risk of **bleeding**.
- **Platelet transfusion**: Indicated when platelet count is very low or if there are **severe bleeding manifestations**.
- **Blood transfusion**: For severe **hemorrhage** or **shock**.
- **Symptomatic treatment**:
  - **Pain relief** for headache and muscle pain.
  - **Anti-nausea medications** as required.
- **Preventing shock**: Monitoring for **dengue shock syndrome (DSS)**. If shock occurs, **aggressive fluid resuscitation** is needed.
- **Hospitalization**: Required for severe cases, especially with **shock** or **organ failure**.

Prevention:

- **Vector control**:
  - Eliminate mosquito breeding sites (standing water).
  - Use **mosquito nets**, **repellents**, and wear **protective clothing**.
  - **Indoor spraying** with insecticides.
- **Vaccine**:
  - **Dengvaxia** (the only dengue vaccine) is recommended for individuals who have had a **previous dengue infection**.
  - **Vaccination in endemic areas** is recommended for those aged 9-45 years who have had prior dengue infection.
- **Personal protection**: Avoid mosquito bites through the use of **mosquito nets**, **repellents**, and **long sleeves**.

Key Points:

- **Dengue Hemorrhagic Fever (DHF)** is a **severe form** of dengue with **bleeding manifestations** and potential for **shock** and **organ failure**.

- **Thrombocytopenia**, **plasma leakage**, and **high hematocrit** are key diagnostic features.
- Early detection and **fluid management** are crucial for **improving survival**.
- **Prevention** includes **mosquito control** and **vaccination** in endemic regions.

Q15. Cerebral malaria.

Severe Falciparum Malaria

- **Most dangerous form** of malaria, responsible for **majority of deaths**.
- **Cerebral malaria** is the **most serious complication** of falciparum malaria.

Management of Severe Falciparum Malaria

- **Hospitalization** is necessary for all severe cases.
- **Prompt treatment** is crucial to prevent life-threatening complications.

Antimalarial Chemotherapy

- **Appropriate antimalarial drugs** should be administered immediately:
  - **Intravenous quinine** or **artesunate** is commonly used.
  - These drugs are essential in managing **severe falciparum malaria** and reducing **mortality**.
- **Follow-up with oral antimalarials**: After intravenous therapy, oral therapy may be continued to eliminate the parasite.

Detection and Treatment of Complications

- **Fluid management**:
  - Monitor and correct **fluid balance** carefully to avoid **overhydration** or **dehydration**.
  - **Intravenous fluids** (e.g., saline, Ringer's lactate) should be given to maintain **circulatory volume**.
- **Electrolyte and acid-base abnormalities**:
  - Monitor and treat **electrolyte imbalances** (e.g., **hypokalemia**, **hyponatremia**) and **acidosis**.
  - Correct **lactic acidosis** if present, as it is a common finding in severe malaria.
- **Management of shock**:
  - **Shock** is a significant complication and needs aggressive **fluid resuscitation**.
  - **Vasopressors** may be required in cases of **septic shock**.

Management of Cerebral Malaria

- **Cerebral malaria** is the most **life-threatening complication** of falciparum malaria and requires:
  - **Close monitoring** of **neurological status**.
  - **Antimalarial therapy** must be continued, as it is critical to reduce the parasite load.
  - **Supportive care** such as **mechanical ventilation** may be required if respiratory failure occurs.

Other Complications

- **Severe anemia**:
  - **Blood transfusion** may be required to treat **severe anemia** caused by **hemolysis**.
- **Acute renal failure**:
  - **Dialysis** may be needed if **renal failure** occurs due to **severe malaria.**
- **Severe hypoglycemia**:
  - **Glucose monitoring** is important as **severe malaria** can lead to **hypoglycemia**, especially with quinine treatment.
- **DIC (Disseminated Intravascular Coagulation)**:
  - Requires **blood product support** and careful management of **coagulation abnormalities**.

Key Points:

- **Severe falciparum malaria** is the most **dangerous** and can cause **death** if not treated aggressively.
- Early diagnosis and **antimalarial therapy** are essential to manage complications such as **cerebral malaria**, **shock**, and **renal failure**.
- **Fluid management**, **electrolyte correction**, and **supportive care** play a critical role in improving **survival**.